

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(rvest)
library(here)
library(cowplot)
here()
```

```
## [1] "/home/guest/EDE_Spring2026_personal"
```

```
mytheme <- theme_gray() +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024

- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: `https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024`

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
the_website <-
  read_html("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024")
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
the_system_name <- the_website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>% html_text()

the_contact_person <- the_website %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>% html_text()

the_ownership <- the_website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>% html_text()

maximum_day_use <- the_website %>%
  html_nodes("th~ td+ td") %>% html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

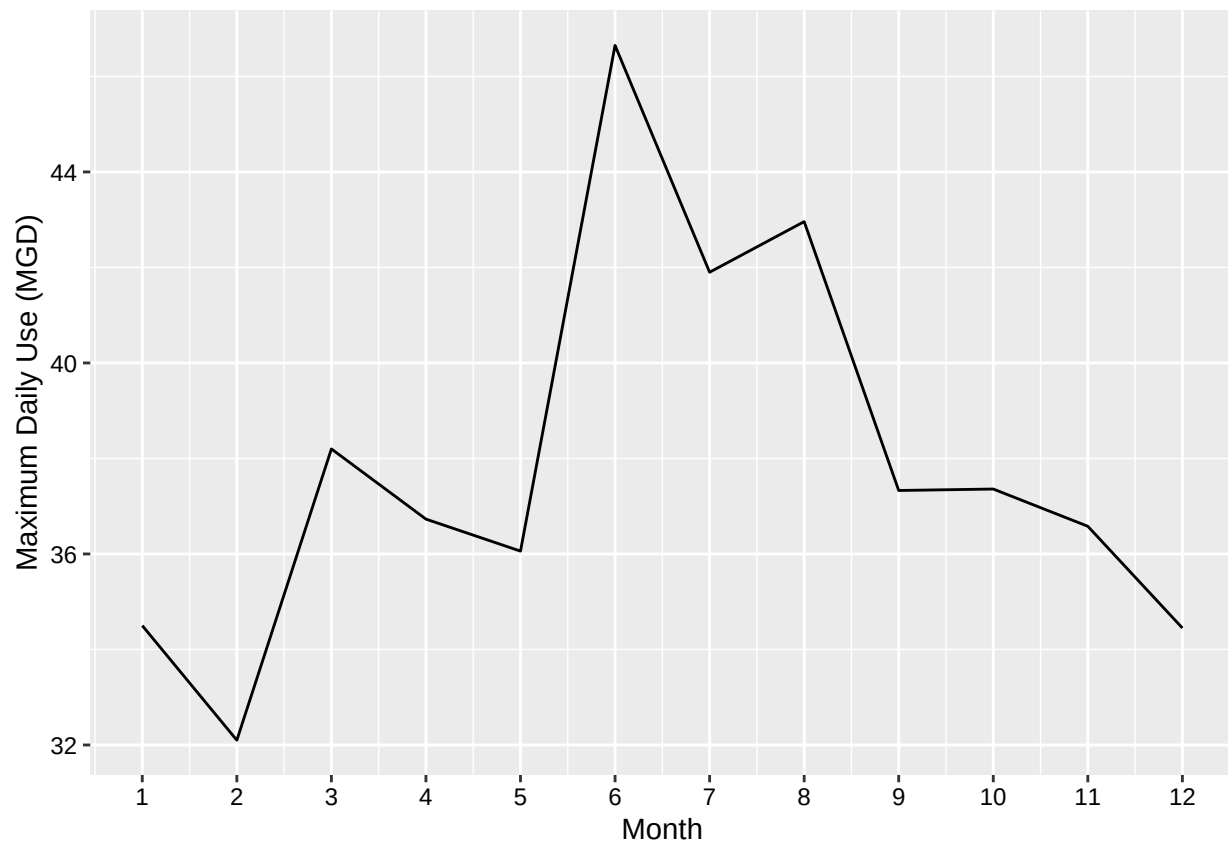
NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4
max_day_df <- data.frame(
  "contact.person" = the_contact_person,
  "ownership" = the_ownership,
  "system.name" = the_system_name,
  "max.day.use.MGD" = as.numeric(maximum_day_use),
  "month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "year" = 2024
)

max_day_df <- max_day_df %>%
  mutate(date = my(paste(month,"-",year)))

#5
max_day_plot <- ggplot(max_day_df, aes(x = month,y=max.day.use.MGD)) +
  geom_line() + #mytheme +
  labs(x="Month", y="Maximum Daily Use (MGD)") +
  scale_x_continuous(
    limits = c(1, 12),
    breaks = 1:12
  )
print(max_day_plot)
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
scrape.it <- function(PWSID, year){
  base_url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid='
  scrape_url <- paste0(base_url, PWSID, "&year=", year)
  the_website <- read_html(scrape_url)

  #setting element address variables
  the_contact_person <- 'div+ table tr:nth-child(4) td:nth-child(2)'
  the_ownership <- 'div+ table tr:nth-child(2) td:nth-child(4)'
  the_system_name <- 'div+ table tr:nth-child(1) td:nth-child(2)'
  maximum_day_use <- 'th~ td+ td'
  #scraping the data items
  the_contact_person <- the_website %>% html_nodes(the_contact_person) %>% html_text()
  the_ownership <- the_website %>% html_nodes(the_ownership) %>% html_text()
  the_system_name <- the_website %>% html_nodes(the_system_name) %>% html_text()
  maximum_day_use <- the_website %>% html_nodes(maximum_day_use) %>% html_text()
  #creating data frame
  df_day_use <- data.frame(
    contact.person = the_contact_person,
    ownership = the_ownership,
    system.name = the_system_name,
    max.day.use.MGD = maximum_day_use,
    month = c(1,5,9,2,6,10,3,7,11,4,8,12),
    PWSID = PWSID,
    year = year
  )

  return(df_day_use)
}
```

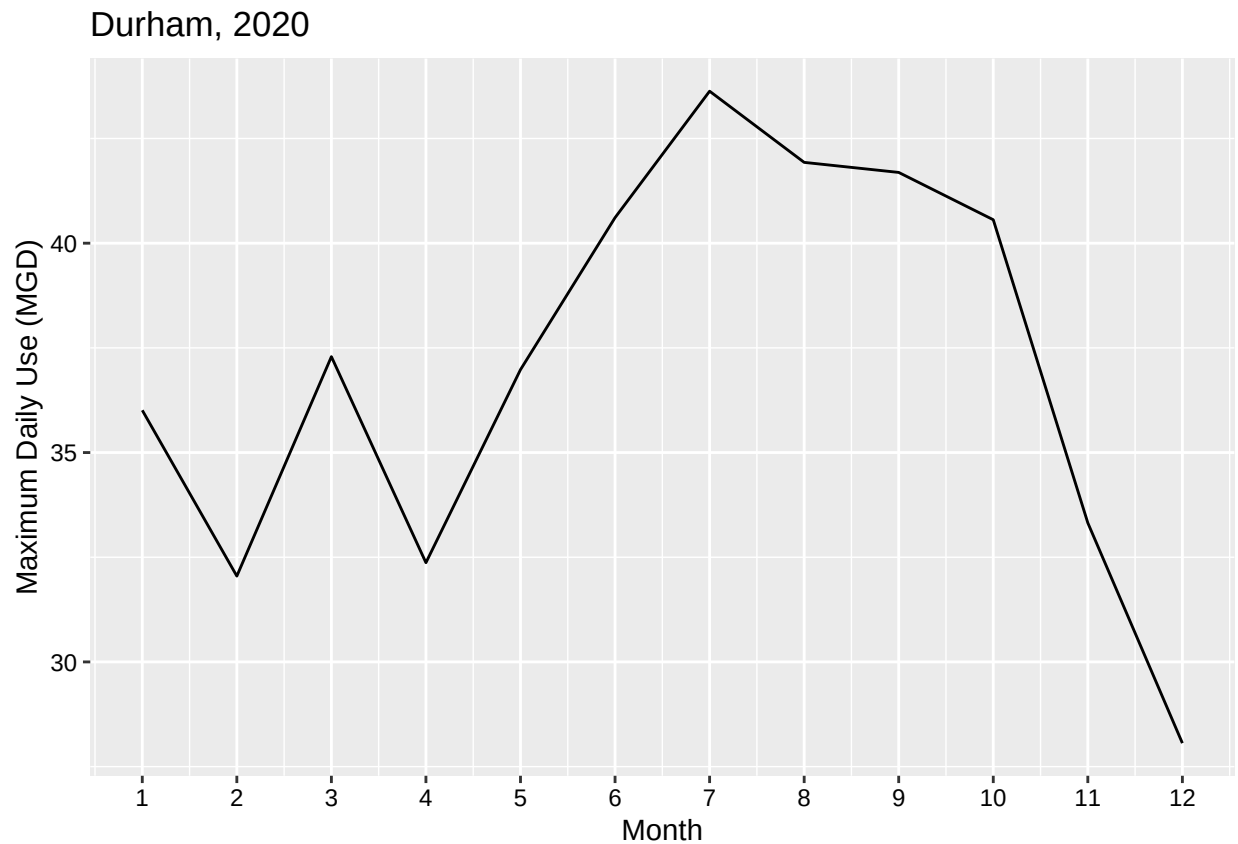
7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID=‘03-32-010’) for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
df <- scrape.it("03-32-010", 2020)

#making sure the max use value and the month are read as numeric and not characters
df <- df %>%
  mutate(
    max.day.use.MGD = as.numeric(gsub(",", "", max.day.use.MGD)),
    month = as.numeric(month)
  )
```

```
df <- df %>%
  mutate(date = my(paste(month,"-",year)))

day_use_plot <- ggplot(df, aes(x = month,y=max.day.use.MGD, group = 1)) +
  geom_line() +
  labs(x="Month", y="Maximum Daily Use (MGD)", title="Durham, 2020") +
  scale_x_continuous(
    limits = c(1, 12),
    breaks = 1:12
  )
print(day_use_plot)
```



```
str(df)
```

```
## 'data.frame': 12 obs. of 8 variables:
## $ contact.person : chr "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ system.name : chr "Durham" "Durham" "Durham" "Durham" ...
## $ max.day.use.MGD: num 36 37 41.7 32 40.6 ...
## $ month : num 1 5 9 2 6 10 3 7 11 4 ...
## $ PWSID : chr "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ date : Date, format: "2020-01-01" "2020-05-01" ...
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data

with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
df2 <- scrape.it("03-92-020",2020)

#making sure the max use value and the month are read as numeric and not characters
df2 <- df2 %>%
  mutate(
    max.day.use.MGD = as.numeric(gsub(",", "", max.day.use.MGD)),
    month = as.numeric(month)
  )

df2 <- df2 %>%
  mutate(date = my(paste(month,"-",year)))

day_use_plot2 <- ggplot(df2, aes(x = month,y=max.day.use.MGD, group = 1)) +
  geom_line() +
  labs(x="Month", y="Maximum Daily Use (MGD)", title="Cary, 2020") +
  scale_x_continuous(
    limits = c(1,12),
    breaks = 1:12
  )
  )

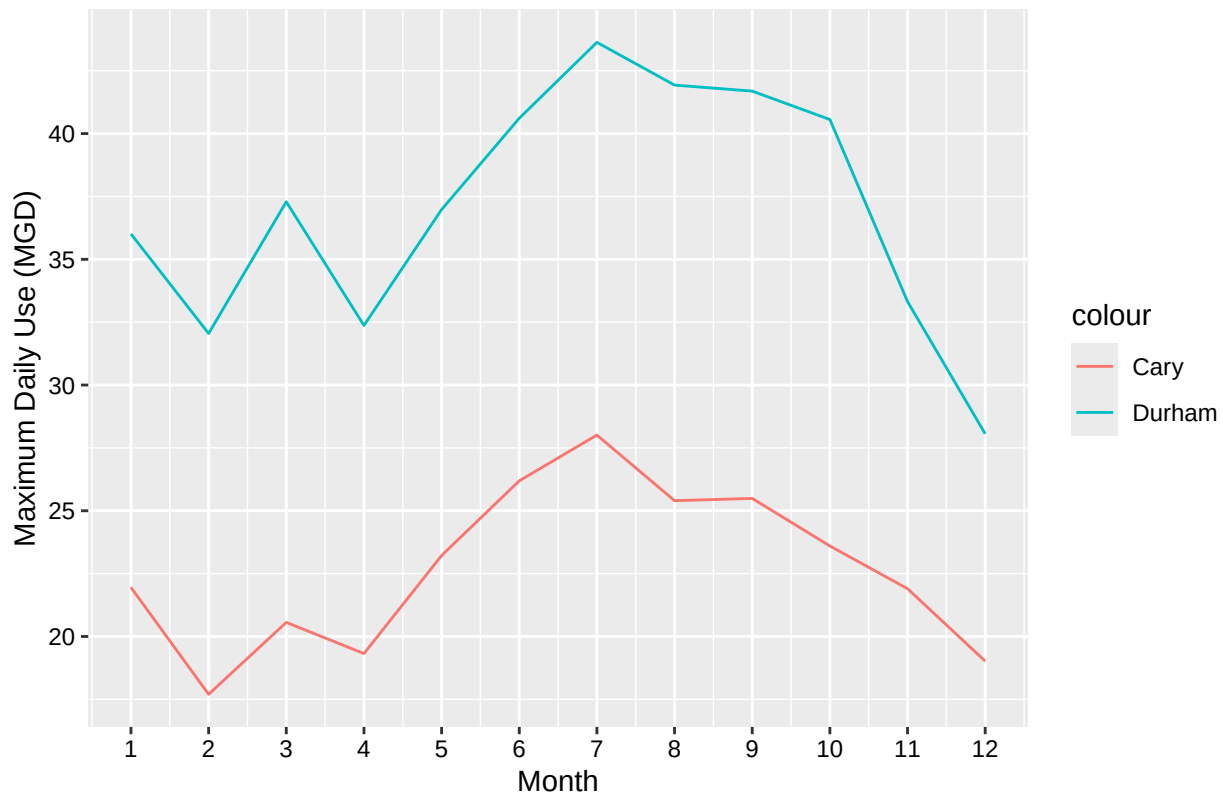
str(df2)
```

```
## 'data.frame': 12 obs. of 8 variables:
## $ contact.person : chr "Jamie Revels" "Jamie Revels" "Jamie Revels" "Jamie Revels" ...
## $ ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ system.name : chr "Cary" "Cary" "Cary" "Cary" ...
## $ max.day.use.MGD: num 21.9 23.2 25.5 17.7 26.2 ...
## $ month : num 1 5 9 2 6 10 3 7 11 4 ...
## $ PWSID : chr "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
## $ year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ date : Date, format: "2020-01-01" "2020-05-01" ...
```

```
compare <- ggplot() +
  geom_line(
    data=df, aes(x = month,y=max.day.use.MGD, group=1, color = "Durham")) +
  geom_line(
    data=df2, aes(x = month,y=max.day.use.MGD, group=1, color="Cary")) +
  labs(
    x="Month", y="Maximum Daily Use (MGD)",
    title = "Water Withdrawl for Cary and Durham, NC") +
  scale_x_continuous(limits = c(1,12), breaks = 1:12) +
  theme(legend.position = "right")

print(compare)
```

Water Withdrawal for Cary and Durham, NC



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
the_years <- 2018:2023
cary_id <- '03-92-020'
cary_withdrawals <- map2(rep(cary_id, length(the_years)), the_years, scrape.it) %>%
  bind_rows()

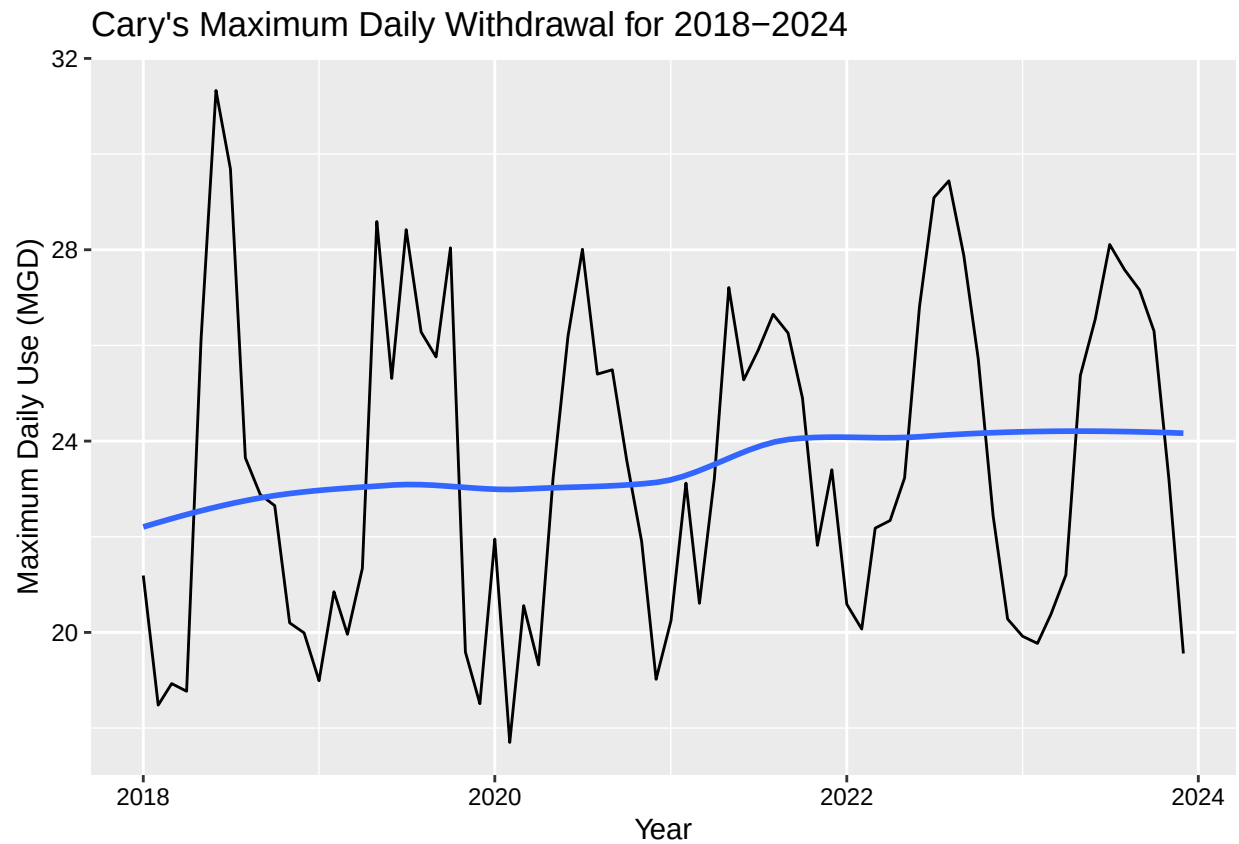
#making sure the max daily use values are read as numeric and not characters
cary_withdrawals <- cary_withdrawals %>%
  mutate(max.day.use.MGD = as.numeric(max.day.use.MGD), month=as.numeric(month))

cary_withdrawals <- cary_withdrawals %>%
  mutate(date = my(paste(month, "-", year)))

cary_plot <- ggplot(cary_withdrawals, aes(x = date, y=max.day.use.MGD)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  labs(x="Year", y="Maximum Daily Use (MGD)",
```

```
title = "Cary's Maximum Daily Withdrawal for 2018-2024")  
print(cary_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: There looks to be a slight increase over time in water used, as the smoothed line is increasing somewhat as time goes on. However, this increase is very slight. >